Tanzania Tourism Expenditure Prediction

Executive Machine Learning Project Report

4,809 records analysed to

predict the total travel costs

Travel to Tanzania

With family, friends, colleagues or alone

Main activities

Wildlife tourism

Cultural tourism

Beach tourism

Tour arrangement

Package tour

- Wildlife & beach
- Detailed itineraries

Independent tour

- Cultural immersion
- Beach

Stay duration

1 day trip

3 days trip

6 days trip

- Small group
- Medium group
- 13 days trip

Budget ?

The Business Challenge

A Pillar of the Economy

- Contributes ~17% of Tanzania's GDP
- 25% of foreign exchange revenues
- Employs over 600,000 people directly
- "Whale" tourists (spending 50M-90M TZS) create extreme variation, making prediction difficult for standard tools.

Executive Impact

-40%

Error Reduction

Surpassing Baseline Performance

The best-performing model (XGBoost) reduces prediction error significantly compared to the naive approach of guessing the average spend.

The model now explains 39.5% of the total variation in tourist spending, enabling more accurate revenue forecasting.

Our Rigorous 6-Step Pipeline

Step 1
Clean Data
Fixed typos & date errors in 4,809 records.

Step 2
Split
70% Train, 15% Val, 15% Sealed Test set.

Step 3
Engineer
Log-transformed spend to handle outliers.

Step 4
Train
Ridge, Random Forest, & XGBoost models.

Step 5
Tune
Systematic hyper-parameter optimization.

Step 6
Explain
SHAP analysis for full model auditability.

"The test set was sealed until the very end to guarantee the final score is honest."

Machine Learning Model Selection

Ridge Regression

A linear formula approach. Strong baseline but struggles with complex non-linear relationships in tourist behavior.

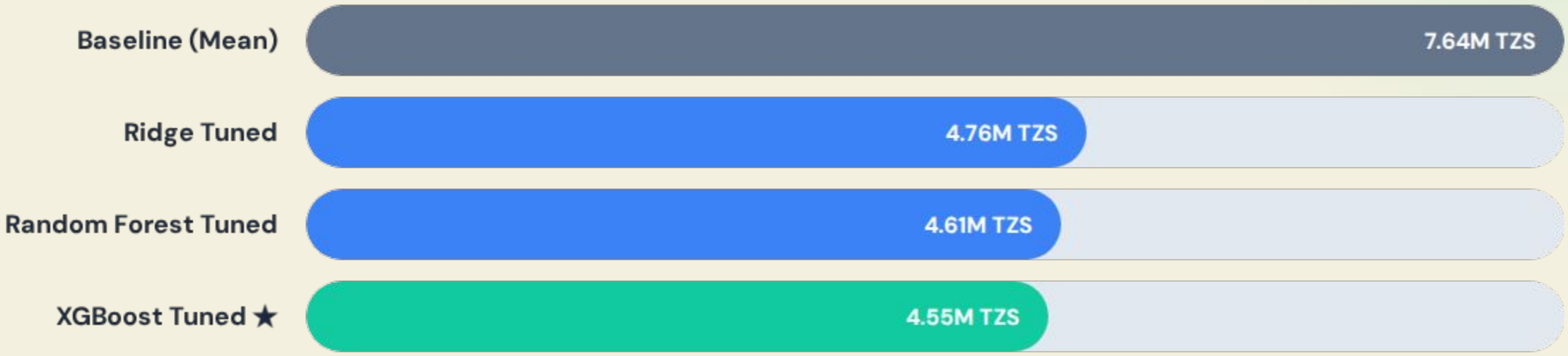
Random Forest

Hundreds of independent decision trees. Improved accuracy by capturing interactions between travel factors.

XGBoost Tuned ★

The winner. Sequential trees that learn from previous errors. Lowest error rate (4.55M TZS MAE) on unseen data.

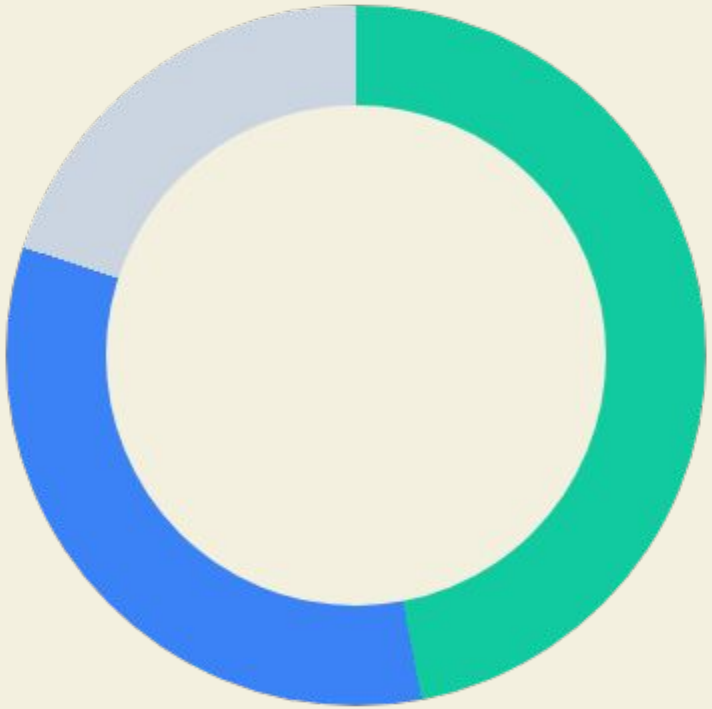
Model Performance (Mean Absolute Error)



Lower MAE is better. The XGBoost model represents a 40.4% improvement over guessing the mean.

All spend values are in Tanzanian Shillings (TZS) – 1,000,000 TZS ≈ 340–350 EUR at current rates.

How Often is the Model Accurate?



47% Predicted within 50% of truth

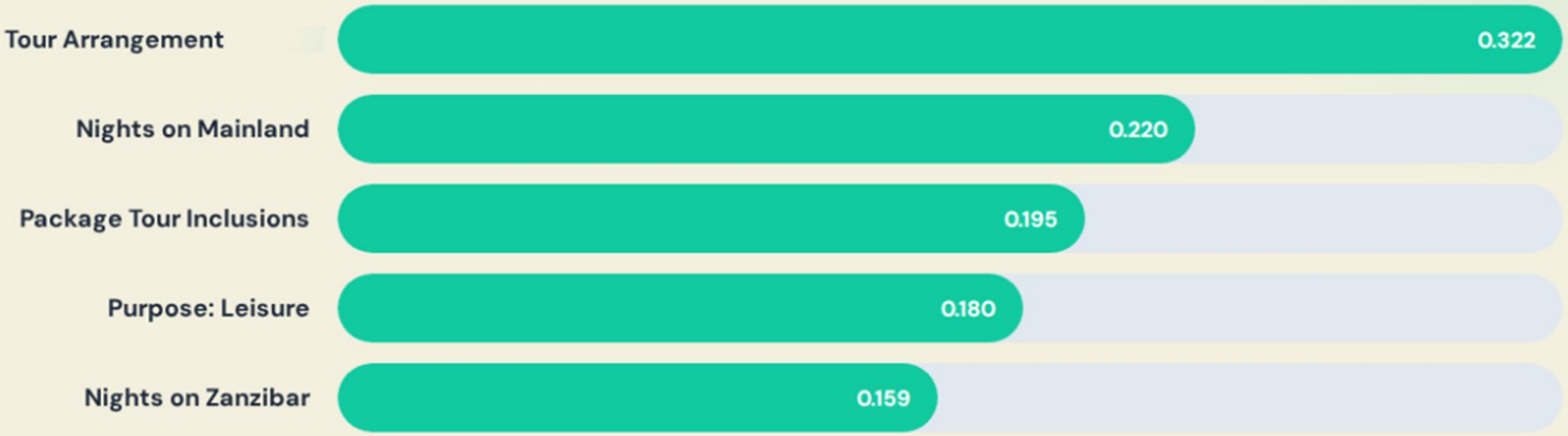
33% Predicted within 50–100% of truth

20% Errors > 100% (mostly luxury)

80% of tourists are predicted within ±100% of their actual spend.

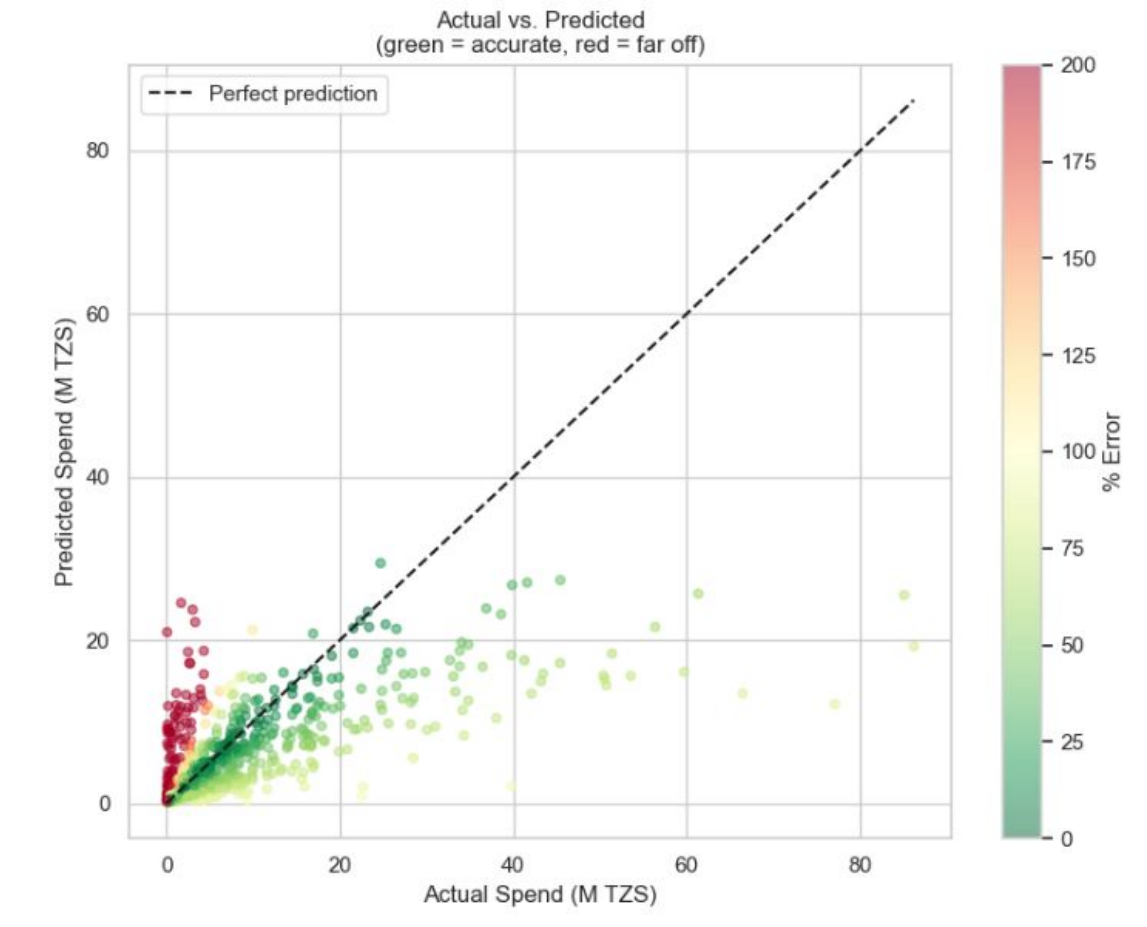
"About half the predictions are accurate and about 20% are poor (mainly for luxury tourists)."

What Drives Tourist Spending?



Surprising Finding: Country of origin that is usually the focus of marketing is not in the top 5. Trip type and duration are much more predictive.

Where does the model work (and where it struggles)?



The models works well for “normal” tourists but struggles with “expensive” tourists

The Model in Action: 3 Profiles

Budget Traveller

Kenya • Solo • 2 nights
Independent • Visiting friends

Prediction Error: 0.4%

Mid-Range Tourist

Germany • Couple • 16 nights
Independent • Leisure

Prediction Error: 1.3%

Premium Safari

USA • Couple • 11 nights
Full Package • Leisure

Prediction Error: 0.1%

The model accurately forecasts spend for diverse tourist types, from budget to luxury safari.

The Model in Action 2

Inaccurate Example #1

South Africa• Solo • 9 nights on midland • 11 nights on Zanzibar • Independent • Leisure

Prediction Error: 53.1 %

Inaccurate Example #2

Malaysia • Solo • 6 nights
Independent • Leisure

Prediction Error: 16.3 %

Inaccurate Example #3

USA • Solo • 11 nights
Full Package • Leisure

Prediction Error: 68.0%

Strategic Recommendations

Objective: Develop a model to predict tourist expenditure to identify high-value segments and optimize marketing spend.

Shift Marketing Lens

Redesign marketing segments around **Trip Type** (Independent vs Package) rather than just Country of Origin.

Prioritize Stay Extension

Duration is a massive spend driver. Promoted itineraries combining **Mainland Safari + Zanzibar** have the highest ROI.

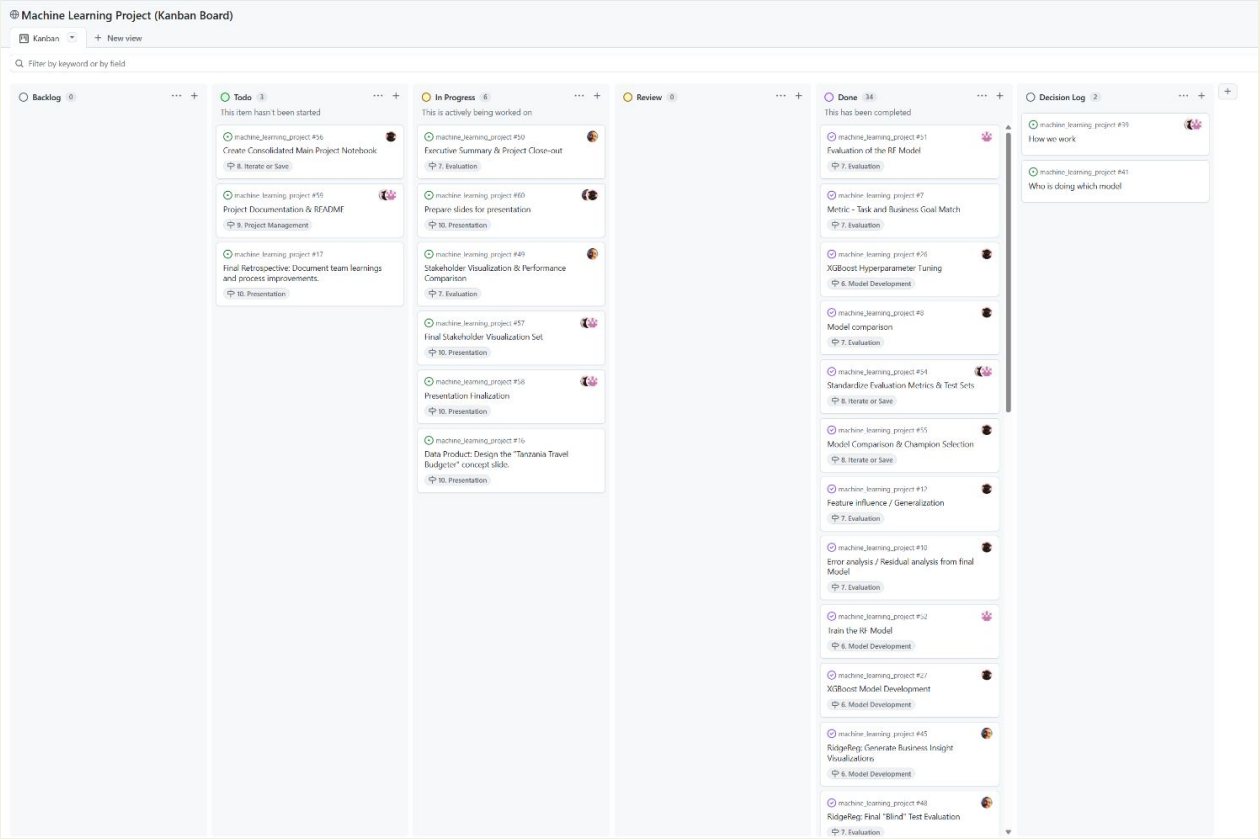
Live Revenue Forecasting

We recommend integrating this logic directly into tour operator booking systems via API or as an extension/add-on.

- Real-time spend estimates upon booking
- Proactive infrastructure & resource planning
- Targeted premium concierge outreach

Agile Project Management

- **Visual Kanban Board:** Set up to track task movement from backlog to deployment.
- **Daily Standups:** Held to synchronize progress and unblock technical issues.
- **Decision Log:** Centralized record of model architecture choices.
- **GitHub Issues:** Assigned tasks to members with dedicated branches.
- **Retrospective:** Planned session to review team workflow efficiency.



Questions?

Thank you for your attention.

Image Sources

https://media.istockphoto.com/id/2219189670/photo/dawn-of-the-wild-serengeti-sunrise.jpg?s=612x612&w=0&k=20&c=RGE2Jhy8TFIBtjKKOFOo9TcI2B077Wwavbw3vO3pqjk=

Source: www.istockphoto.com

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Source: pngtree.com

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